

Recursion Extra

Exercise 1 - Sentence reverse

Write a program that is able to take a sentence and give back its reversed version.

Implement it first using an **iterative** approach and then using a **recursive** approach.

Exercise 2 - Recursive scale

Develop a program that solves the following problem:

You have 9 balls: 8 balls have the same weight, and 1 ball is heavier (it weighs 1 gram more than the others).

You also have access to a two-pan balance scale.

How can you determine which ball is the heavier one using the fewest possible weighings?

Implement a program that simulates the process and returns the index of the heavier ball.

To solve the problem you must take advantage of **recursion**.